Calculating Probabilities

Name _____ Date _____ No calculator Version 1 Scan to practise this topic online

KEY WORDS probability *(tūponotanga)* · outcome *(putanga)* · event *(takahanga)* · equally likely *(tūpono ōrite)*

A The probability of an outcome ●○○

Remember: Probability $= \dfrac{\text{number of outcomes you want}}{\text{total number of outcomes}}$, when every outcome is equally likely. A dice has 6 outcomes, so the probability of rolling a 4 is $\dfrac{1}{6}$.

1. A fair dice is rolled. Write the probability of each event as a fraction in its simplest form.

a) Rolling a 4 _____ **b)** Rolling an even number _____

c) Rolling a number greater than 4 _____ **d)** Rolling a 7 _____

e) Rolling a number less than 7 _____ **f)** Rolling a 1 or a 2 _____

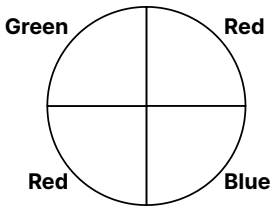
2. A bag holds 3 red, 5 blue and 2 green counters. One counter is taken out without looking. Write the probability of each event as a fraction in its simplest form.

a) A red counter _____ **b)** A blue counter _____ **c)** A green counter _____

B Spinners ●●○

Watch out: Count the parts, not the colours. A spinner with 4 equal parts, 2 of them red, lands on red with probability $\dfrac{2}{4} = \dfrac{1}{2}$.

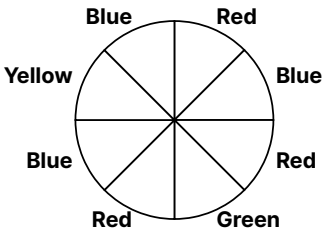
3. This spinner has 4 equal parts. Find the probability that it lands on:



a) red _____ **b)** blue _____

c) green _____ **d)** yellow _____

4. This spinner has 8 equal parts. Find the probability that it lands on:



a) red _____ **b)** blue _____

c) green _____ **d)** yellow _____

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C Combinations of outcomes ●●○

⚠ **Watch out:** Count every outcome that fits the event. For "a number less than 4" on a spinner numbered 1 to 8, the outcomes 1, 2 and 3 all count.

5. A fair spinner is numbered 1 to 8. Find the probability of each event, as a fraction in its simplest form.

a) A number less than 4 **b)** An even number **c)** A 1 or an 8

d) A number greater than 5 **e)** A multiple of 3 **f)** A number greater than 8

6. A bag holds 5 red, 4 blue, 2 green and 1 yellow counters. One counter is taken out without looking. Find the probability that it is:

a) red or blue **b)** green or yellow **c)** not red

D Probabilities add to 1 ●●●

💡 **Remember:** An event either happens or it does not, so the two probabilities add to 1. If the probability of rain is $\frac{3}{10}$, the probability of no rain is $1 - \frac{3}{10} = \frac{7}{10}$.

7. The probability that an event happens is given. Find the probability that it does not happen.

a) $\frac{3}{8}$ **b)** $\frac{1}{6}$ **c)** $\frac{2}{5}$ **d)** $\frac{7}{10}$

e) $\frac{1}{2}$ **f)** $\frac{3}{4}$ **g)** $\frac{5}{9}$ **h)** $\frac{1}{100}$

8. Answer each question.

a) The probability that a bus is late is $\frac{1}{5}$. What is the probability that it is not late?

b) A spinner lands on red with probability $\frac{3}{8}$. What is the probability that it does not land on red?

E Comparing likelihood ●●●

9. Which is more likely? Show your working.

a) rolling a 6 on a fair dice, or getting a head on a fair coin?

b) picking a red counter from a bag with 1 red and 3 blue counters, or picking a red counter from a bag with 3 red and 7 blue counters?

c) picking a vowel from the letters of MATHS, or picking a vowel from the letters of NUMBER?

★ Extension

10. A bag holds 10 counters. The probability of picking a red counter is $\frac{2}{5}$. How many red counters are in the bag?

11. A spinner has 3 parts coloured red, blue and green. Tama says, "The probability of red must be one third." When would he be wrong?

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Calculating Probabilities

Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

KEY WORDS probability *(tūponotanga)* · outcome *(putanga)* · event *(takahanga)* · equally likely *(tūpono ōrite)*

A The probability of an outcome ●○○

Remember: Probability $= \dfrac{\text{number of outcomes you want}}{\text{total number of outcomes}}$, when every outcome is equally likely. A dice has 6 outcomes, so the probability of rolling a 4 is $\dfrac{1}{6}$.

1. A fair dice is rolled. Write the probability of each event as a fraction in its simplest form.

a) Rolling a 3 _____ **b)** Rolling an even number _____

c) Rolling a number greater than 2 _____ **d)** Rolling a 7 _____

e) Rolling a number less than 7 _____ **f)** Rolling a 1 or a 2 _____

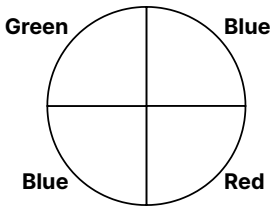
2. A bag holds 4 red, 5 blue and 1 green counters. One counter is taken out without looking. Write the probability of each event as a fraction in its simplest form.

a) A red counter _____ **b)** A blue counter _____ **c)** A green counter _____

B Spinners ●●○

Watch out: Count the parts, not the colours. A spinner with 4 equal parts, 2 of them red, lands on red with probability $\dfrac{2}{4} = \dfrac{1}{2}$.

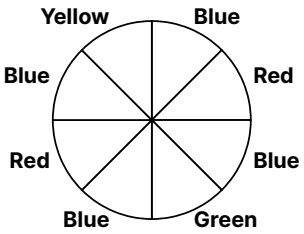
3. This spinner has 4 equal parts. Find the probability that it lands on:



a) blue _____ **b)** red _____

c) green _____ **d)** yellow _____

4. This spinner has 8 equal parts. Find the probability that it lands on:



a) blue _____ **b)** red _____

c) green _____ **d)** yellow _____

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C Combinations of outcomes ●●○

⚠ **Watch out:** Count every outcome that fits the event. For "a number less than 4" on a spinner numbered 1 to 8, the outcomes 1, 2 and 3 all count.

5. A fair spinner is numbered 1 to 8. Find the probability of each event, as a fraction in its simplest form.

a) A number less than 3 **b)** An even number **c)** A 1 or an 8

d) A number greater than 6 **e)** A multiple of 3 **f)** A number greater than 8

6. A bag holds 4 red, 3 blue, 3 green and 2 yellow counters. One counter is taken out without looking. Find the probability that it is:

a) red or blue **b)** green or yellow **c)** not red

D Probabilities add to 1 ●●●

Remember: An event either happens or it does not, so the two probabilities add to 1. If the probability of rain is $\frac{3}{10}$, the probability of no rain is $1 - \frac{3}{10} = \frac{7}{10}$.

7. The probability that an event happens is given. Find the probability that it does not happen.

a) $\frac{5}{8}$ **b)** $\frac{5}{6}$ **c)** $\frac{3}{5}$ **d)** $\frac{3}{10}$

e) $\frac{1}{2}$ **f)** $\frac{1}{4}$ **g)** $\frac{4}{9}$ **h)** $\frac{99}{100}$

8. Answer each question.

a) The probability that a bus is late is $\frac{1}{4}$. What is the probability that it is not late?

b) A spinner lands on red with probability $\frac{5}{8}$. What is the probability that it does not land on red?

E Comparing likelihood ●●●

9. Which is more likely? Show your working.

a) rolling a 1 on a fair dice, or getting a tail on a fair coin?

b) picking a red counter from a bag with 1 red and 4 blue counters, or picking a red counter from a bag with 3 red and 7 blue counters?

c) picking a vowel from the letters of GRAPH, or picking a vowel from the letters of RANDOM?

★ Extension

10. A bag holds 12 counters. The probability of picking a red counter is $\frac{1}{4}$. How many red counters are in the bag?

11. A spinner has 3 parts coloured red, blue and green. Tama says, "The probability of red must be one third." When would he be wrong?

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